

COMMONWEALTH OF AUSTRALIA

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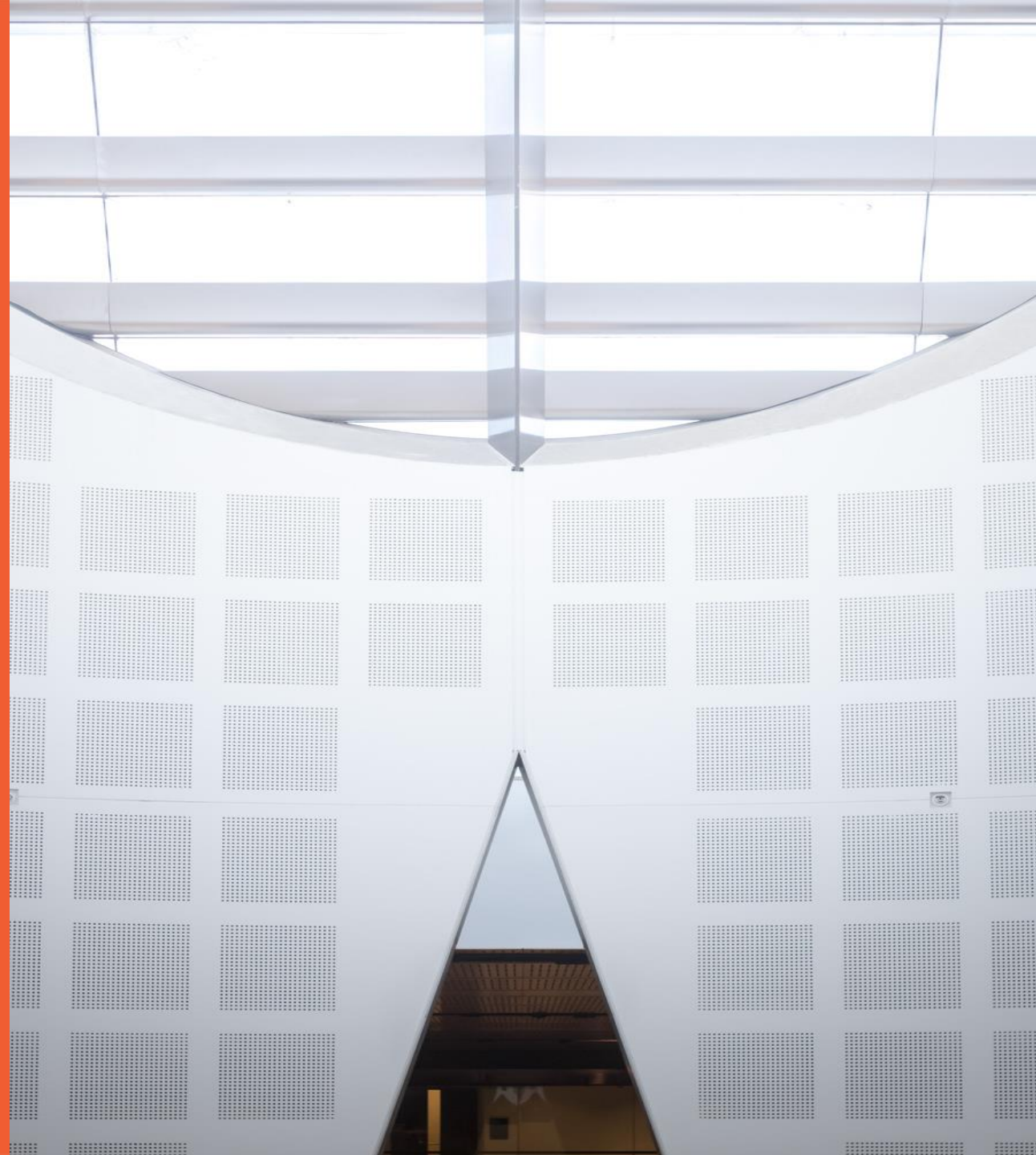
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COMPx270: Randomised and
Advanced Algorithms
Lecture 11: Learning and testing
probability distributions

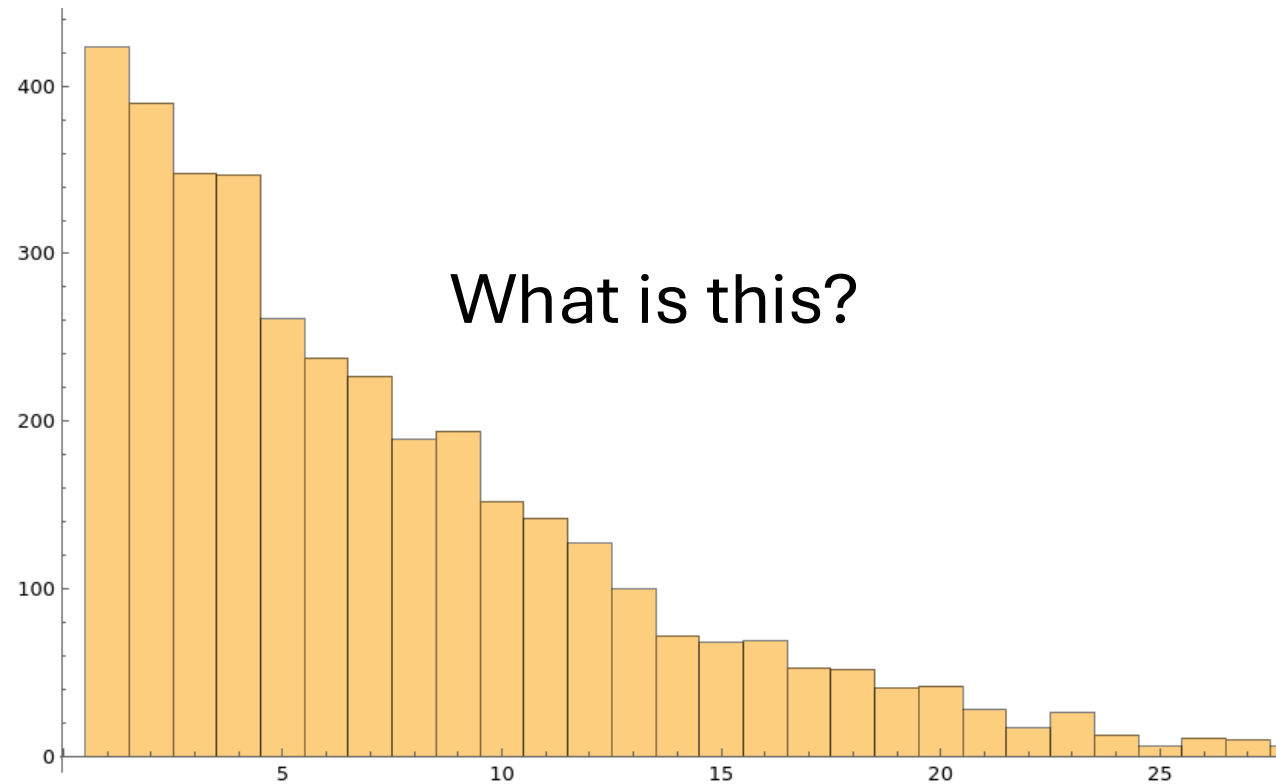
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A question



Learning and testing (discrete) probability distributions

Given* a probability distribution** p over a domain $\mathcal{X}$ of size $k \gg 1$.

- Learn p
- Learn a parameter of p (estimation)
- Learn one bit (test if p satisfies some property)

Preliminaries on probability distributions

Preliminaries on probability distributions

Fact.
$$\text{TV}(p, q) = \frac{1}{2} \sum_{x \in \mathcal{X}} |p(x) - q(x)| = \frac{1}{2} \|p - q\|_1$$

Preliminaries on probability distributions

Data Processing Inequality (DPI). Take any function $f: \mathcal{X} \rightarrow \mathcal{Y}$. For $X \sim p$, $Y \sim q$, let p' be the distribution of $f(X)$, and q' the distribution of $f(Y)$. Then $TV(p', q') \leq TV(p, q)$.

A view of TV distance

Alice and Bob play a game, where they both know two probability distributions $\mathbf{p}, \mathbf{q}$. Alice starts by tossing a fair coin, and does not show the outcome to Bob: if it is Heads, then she draws $x \sim \mathbf{p}$; if it is Tails, she draws $x \sim \mathbf{q}$. Then she shows the value of x to Bob, who must guess if the coin toss was Heads. Clearly, just by random guessing, Bob can win the game with probability $1/2$. What the lemma says is that he can do better: there is a strategy for him to win with probability

$$\Pr[\text{Bob wins}] = \frac{1}{2} + \frac{d_{\text{TV}}(\mathbf{p}, \mathbf{q})}{2}$$

and, moreover, this is the best possible.

The case of a coin

How many times n do you need to flip the coin to learn its true bias p to accuracy $\pm\epsilon$, and be correct with probability at least $1 - \delta$?

The case of a coin

Theorem 50. *Suppose we are promised that the true bias p of the coin satisfies $0 \leq p < q \leq \frac{1}{2}$, for some known value q . Then estimating the bias of the coin to an additive ε , with probability at least $1 - \delta$, can be done with $n = O\left(\frac{q}{\varepsilon^2} \log \frac{1}{\delta}\right)$ i.i.d. samples. (Moreover, this is optimal.)*

The case of a coin

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Corollary 50.1. Estimating the bias of a coin to an additive ε , with probability at least $1 - \delta$, can be done with $n = O\left(\frac{1}{\varepsilon^2} \log \frac{1}{\delta}\right)$ i.i.d. samples. (Moreover, this is optimal.)

The case of a coin

Theorem 50. Suppose we are promised that the true bias p of the coin satisfies $0 \leq p < q \leq \frac{1}{2}$, for some known value q . Then estimating the bias of the coin to an additive ϵ , with probability at least $1 - \delta$, can be done with $n = O\left(\frac{q}{\epsilon^2} \log \frac{1}{\delta}\right)$ i.i.d. samples. (Moreover, this is optimal.)

The case of a coin: what about testing?

"Is my coin fair?"

The case of a coin: what about testing?

Theorem 51. *Testing whether the bias of a coin is $1/2$ or at least $1/2 + \varepsilon$, with probability at least $1 - \delta$, can be done with $n = O\left(\frac{1}{\varepsilon^2} \log \frac{1}{\delta}\right)$ i.i.d. samples. (Moreover, this is optimal.)*

The case of a coin: what about testing?

Theorem 52. For any $0 < \alpha \leq 1/2$ and $\varepsilon \in (0, 1]$, testing whether the bias of a coin is at most α or at least $\alpha(1 + \varepsilon)$, with probability at least $1 - \delta$, can be done with $n = O\left(\frac{1}{\alpha\varepsilon^2} \log \frac{1}{\delta}\right)$ i.i.d. samples.

The case of a coin: what about testing?

Theorem 52. For any $0 < \alpha \leq 1/2$ and $\epsilon \in (0, 1]$, testing whether the bias of a coin is at most α or at least $\alpha(1 + \epsilon)$, with probability at least $1 - \delta$, can be done with $n = O\left(\frac{1}{\alpha\epsilon^2} \log \frac{1}{\delta}\right)$ i.i.d. samples.

Beyond coins: k is large

Domain sizes grow quite fast, and in most settings k is huge.

Learning in TV distance

Unknown p over $\mathcal{X}$ ($|\mathcal{X}| = k \gg 1$).

Parameters ε, δ

Get $x_1, \dots, x_n \sim p$ (for n we choose)

Goal: output an ε -approximation of p with probability at least $1 - \delta$, minimizing the (worst-case) **sample complexity** $n = n(k, \varepsilon, \delta)$.

Learning in TV distance: first attempt

Learning in TV distance: second attempt

Learning in TV distance: third attempt

Theorem 53. *Learning an unknown distribution $\mathbf{p} \in \Delta(k)$ to total variation distance ϵ (with success probability $1 - \delta$) can be done with*

$$n = O\left(\frac{k + \log \frac{1}{\delta}}{\epsilon^2}\right)$$

i.i.d. samples. (Moreover, this is optimal.)

Learning in TV distance: third attempt

Theorem 53. Learning an unknown distribution $\mathbf{p} \in \Delta(k)$ to total variation distance ε (with success probability $1 - \delta$) can be done with

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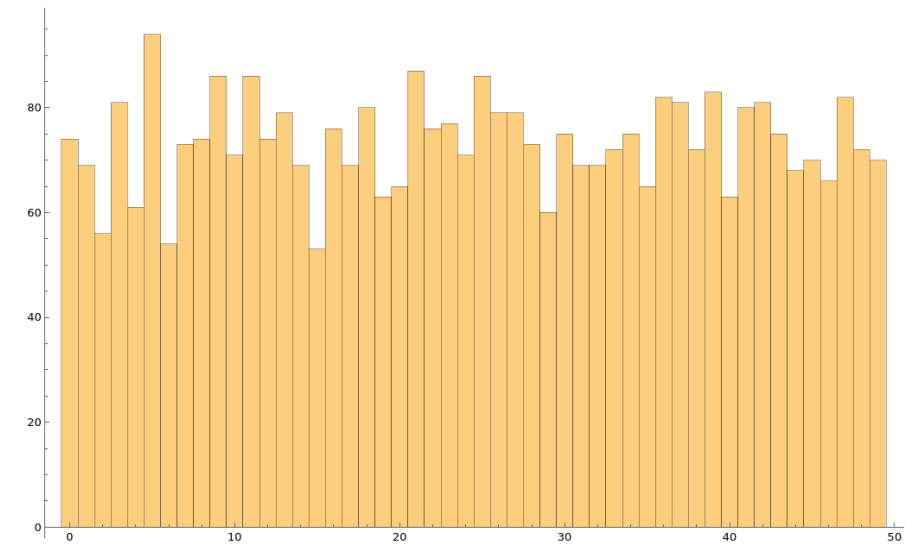
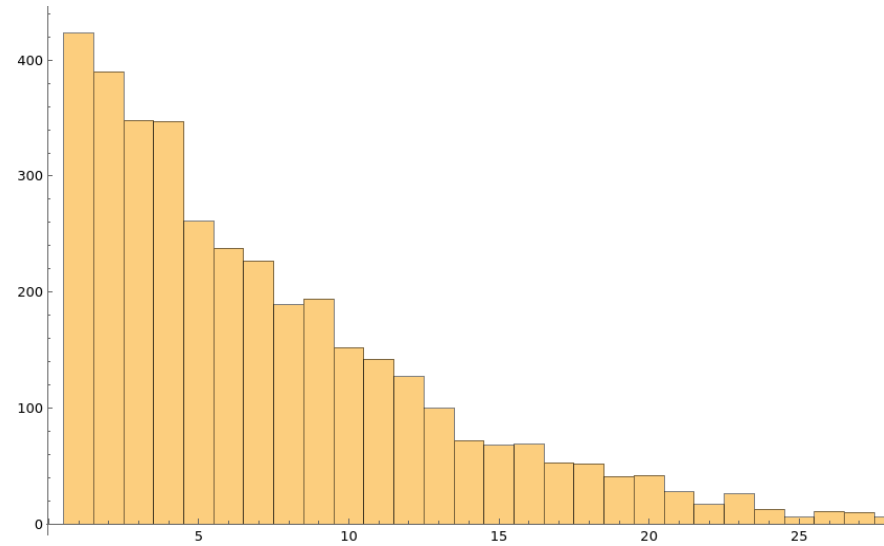
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Testing in TV distance



Testing in TV distance: identity testing

Give an algorithm A which takes parameters $\epsilon, \delta \in (0, 1]$ and n samples from $\mathbf{p}$, and:

- If $\mathbf{p} = \mathbf{q}$, then $\Pr[A \text{ outputs yes}] \geq 1 - \delta$;
- If $d_{\text{TV}}(\mathbf{p}, \mathbf{q}) > \epsilon$, then $\Pr[A \text{ outputs no}] \geq 1 - \delta$

(if $0 < d_{\text{TV}}(\mathbf{p}, \mathbf{q}) \leq \epsilon$, then A is off the hook and can output whatever).

Testing in TV distance: identity testing via learning

Testing in TV distance: uniformity is all you need

Theorem 54 (Identity to uniformity reduction). *Suppose there is an algorithm A for uniformity testing, which takes $n = n(k, \epsilon, \delta)$ i.i.d. samples from the unknown distribution. Then there is an algorithm A' for identity testing over a domain of size k to any fixed $\mathbf{q} \in \Delta(k)$, which takes $n = n(4k, \epsilon/4, \delta)$ i.i.d. samples from the unknown distribution. Moreover, A' is efficient if A is.*

Testing in TV distance: uniformity testing

Testing in TV distance: uniformity testing

Theorem 55. *Testing uniformity of an unknown distribution $\mathbf{p} \in \Delta(k)$ to total variation distance ε (with success probability $2/3$) can be done with*

$$n = O\left(\frac{\sqrt{k}}{\varepsilon^2}\right)$$

i.i.d. samples, using Algorithm 21. (Moreover, this is optimal for constant success probability.)

Testing in TV distance: uniformity testing, key ideas

One idea: use ℓ_2 as a **proxy** for ℓ_1 (TV)!

Key facts.

1. Cauchy-Schwarz:
$$d_{\text{TV}}(\mathbf{p}, \mathbf{u}_k) = \frac{1}{2} \|\mathbf{p} - \mathbf{u}_k\|_1 \leq \frac{\sqrt{k}}{2} \|\mathbf{p} - \mathbf{u}_k\|_2$$

2. "Convenient fact":
$$\|\mathbf{p} - \mathbf{u}_k\|_2^2 = \sum_{i=1}^k (\mathbf{p}(i) - 1/k)^2 = \sum_{i=1}^k \mathbf{p}(i)^2 - 1/k = \|\mathbf{p}\|_2^2 - 1/k$$

Testing in TV distance: uniformity testing, key ideas

(Another) fact.

$$\Pr_{x,y \sim \mathbf{p}} [x = y] = \sum_{i=1}^k \mathbf{p}(i)^2 = \|\mathbf{p}\|_2^2$$

I.e., the squared ℓ_2 norm is the **collision probability**.

Testing in TV distance: uniformity testing, algorithm

Input: Multiset of n i.i.d. samples $x_1, \dots, x_n \in \mathcal{X}$, parameters $\varepsilon \in (0, 1]$ and $k = |\mathcal{X}|$

1: Set $\tau \leftarrow \frac{1+2\varepsilon^2}{k}$

2: Compute

▷ $O(n)$ time if $\mathcal{X}$ is known

$$Z = \frac{1}{\binom{n}{2}} \sum_{1 \leq s < t \leq n} \mathbb{1}_{\{x_s = x_t\}} = \frac{1}{\binom{n}{2}} \sum_{j \in \mathcal{X}} \binom{N_j}{2}$$

where $N_j \leftarrow \sum_{t=1}^n \mathbb{1}_{\{x_t = j\}}$.

3: **if** $Z \geq \tau$ **then return** no

▷ Not uniform

4: **else return** yes

▷ Uniform

Testing in TV distance: uniformity testing

1. Compute expectation

2. Bound variance

3. Conclude via Chebyshev

Input: Multiset of n i.i.d. samples $x_1, \dots, x_n \in \mathcal{X}$, parameters $\epsilon \in (0, 1]$ and $k = |\mathcal{X}|$
1: Set $\tau \leftarrow \frac{1+2\epsilon^2}{k}$
2: Compute $\triangleright O(n)$ time if $\mathcal{X}$ is known

$$Z = \frac{1}{\binom{n}{2}} \sum_{1 \leq s < t \leq n} \mathbb{1}_{\{x_s = x_t\}} = \frac{1}{\binom{n}{2}} \sum_{j \in \mathcal{X}} \binom{N_j}{2}$$

where $N_j \leftarrow \sum_{i=1}^n \mathbb{1}_{\{x_i=j\}}$.
3: **if** $Z \geq \tau$ **then return** no $\triangleright$ Not uniform
4: **else return** yes $\triangleright$ Uniform

Testing in TV distance: uniformity testing, summary

Theorem 55. *Testing uniformity of an unknown distribution $\mathbf{p} \in \Delta(k)$ to total variation distance ε (with success probability $2/3$) can be done with*

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Testing in TV distance: uniformity testing, summary

Theorem 55. *Testing uniformity of an unknown distribution $\mathbf{p} \in \Delta(k)$ to total variation distance ϵ (with success probability $2/3$) can be done with*

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💡 Can do $O\left(\frac{\sqrt{k} \log(1/\delta)}{\epsilon^2}\right)$ by repetition+majority vote, or $O\left(\frac{\sqrt{k \log(1/\delta)} + \log(1/\delta)}{\epsilon^2}\right)$ with a different algo

Summary

- Learning p to TV distance ϵ :

$\Theta(k/\epsilon^2)$ samples

- Testing $p=u$ vs. $TV(p,u) > \epsilon$:

$\Theta(\sqrt{k}/\epsilon^2)$ samples

- Estimating $TV(p,u)$ to $\pm\epsilon$:

$\Theta(k/(\epsilon^2 \log k))$ samples

Want to learn more?

 <https://ccanonne.github.io/survey-topics-dt.html>